

Efficiently Learning Robust Torque-based Locomotion Through Reinforcement with Model-Based Supervision

Yashuai Yan^{1,*}, Tobias Egle^{2,*}, Christian Ott^{2,3}, Dongheui Lee^{1,3}

¹TU Wien Autonomous Systems Lab, ²TU Wien Institute for Automation and Control, ³German Aerospace Center (DLR) Institute for Robotics and Mechatronics, *Contributed equally to the work.



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1 CHALLENGE / MOTIVATION

Introduction

We introduce a **hybrid** approach that combines **model-based and learning-based approaches** that enables **Torque-based bipedal locomotion** that is

- ✓ Natural & agile
- ✓ Energy-efficient
- ✓ Compliant

Comparison

End-2-End RL

RL

learns from data to compute joint positions/torques

Advantage: handles uncertainty and disturbance well, robust

Disadvantage: data-inefficient, limited guidance in training, can be imprecise, noisy and have safety risks

Model-based Control

MB

relies on physical model to compute joint torques

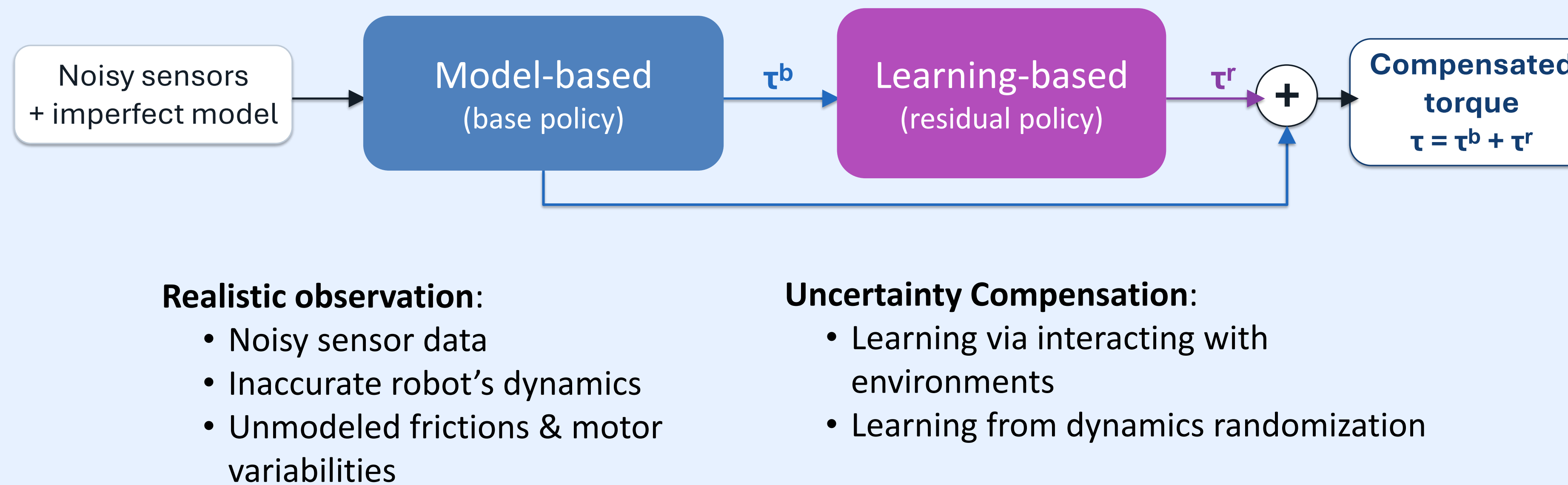
Advantage: precise and smooth actuator commands, high frequency, convergence and safety guaranties

Disadvantage: Sensitive to model-uncertainty and inaccurate/noisy state estimation

Key Idea: Combine MB and RL and learn only what the model cannot predict.

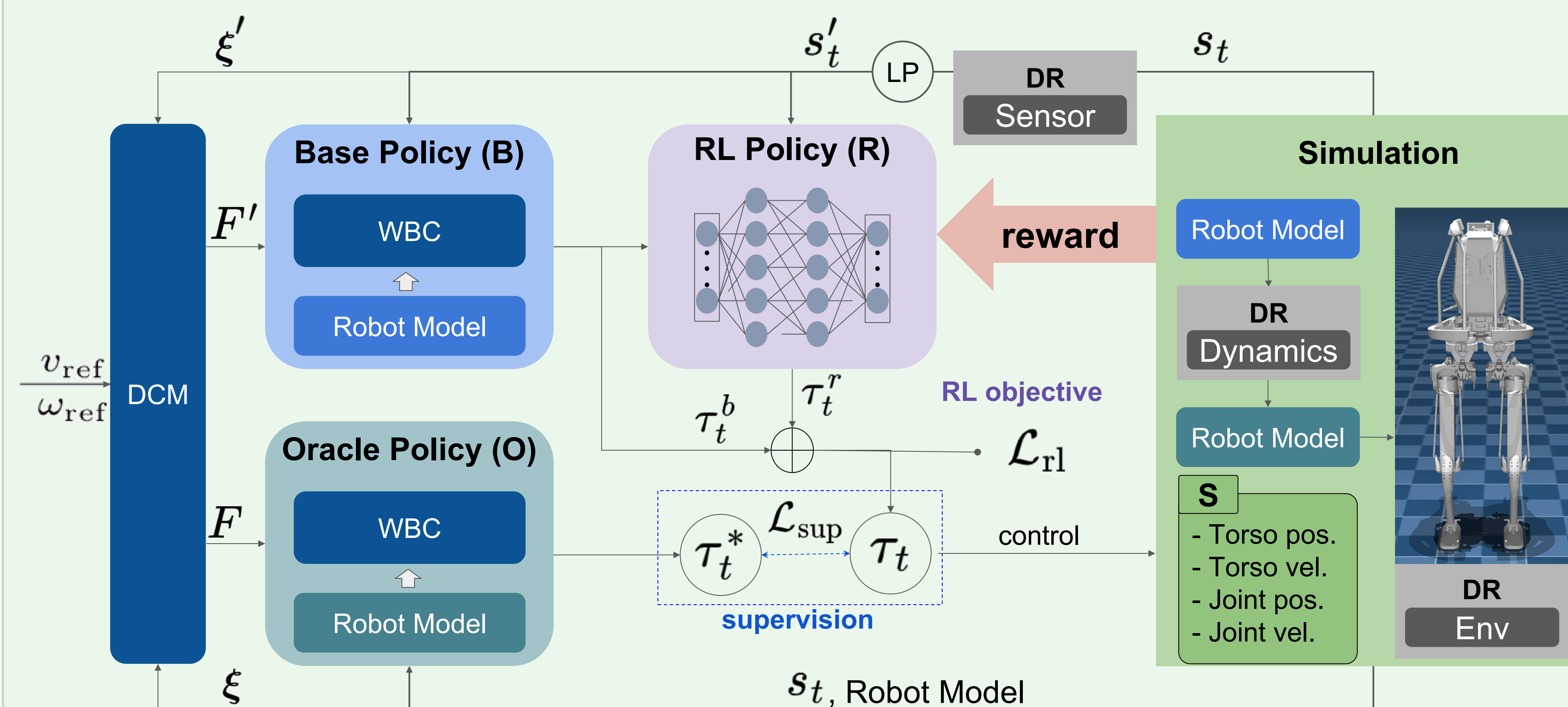
2 HYBRID FRAMEWORK (Model-Based + Residual RL + Oracle Supervision)

Deployment



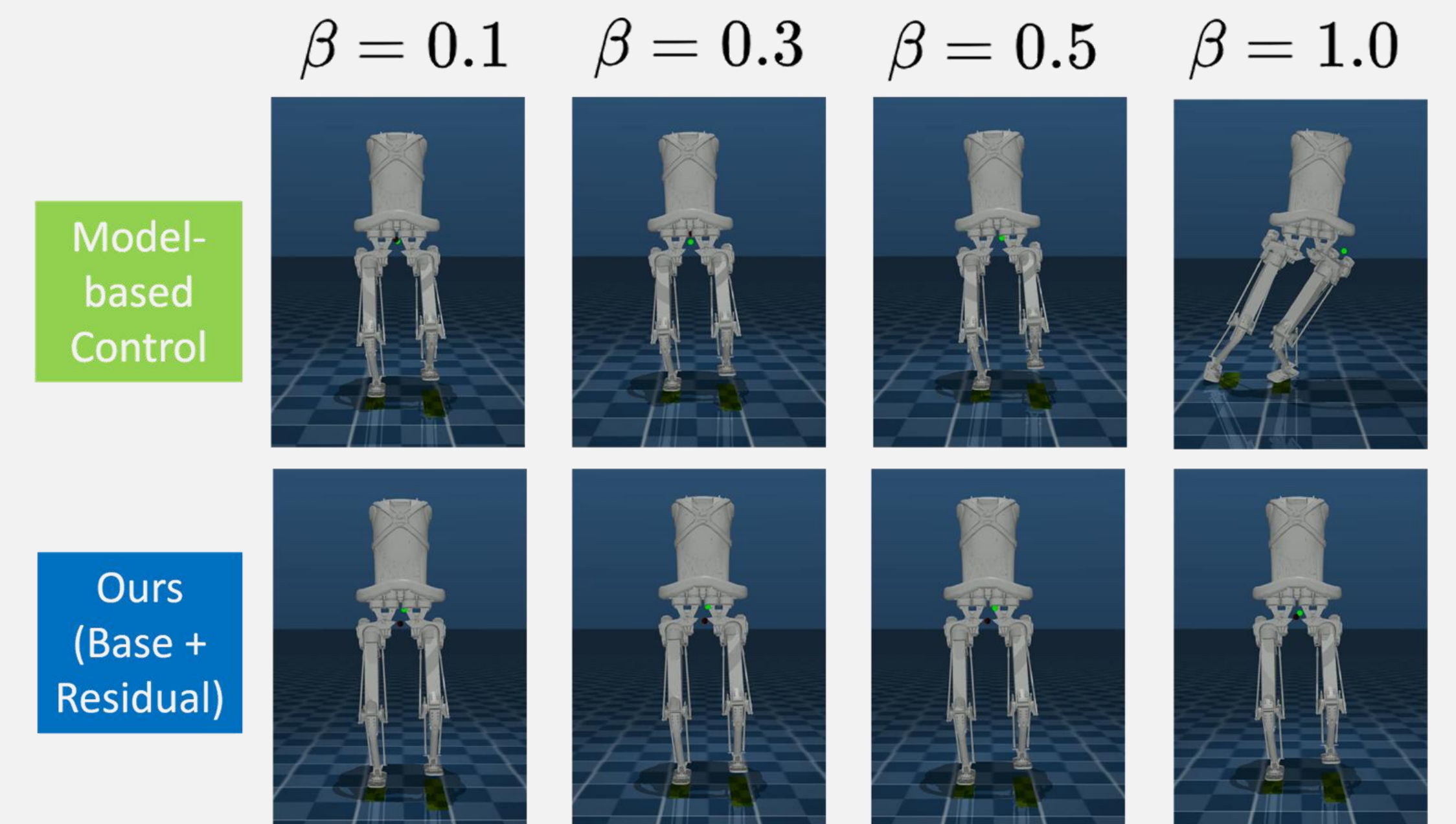
Training

- The **Base policy (B)** is a model-based controller: **noisy** sensor data and **inaccurate** robot model.
- The **Residual policy (R)** learns by reinforcement to maximize the rewards.
- One key component is the **Oracle policy (O)** with **true system states** and **accurate** robot model. The Oracle policy **supervises** the Residual policy with an **additional supervision loss**.



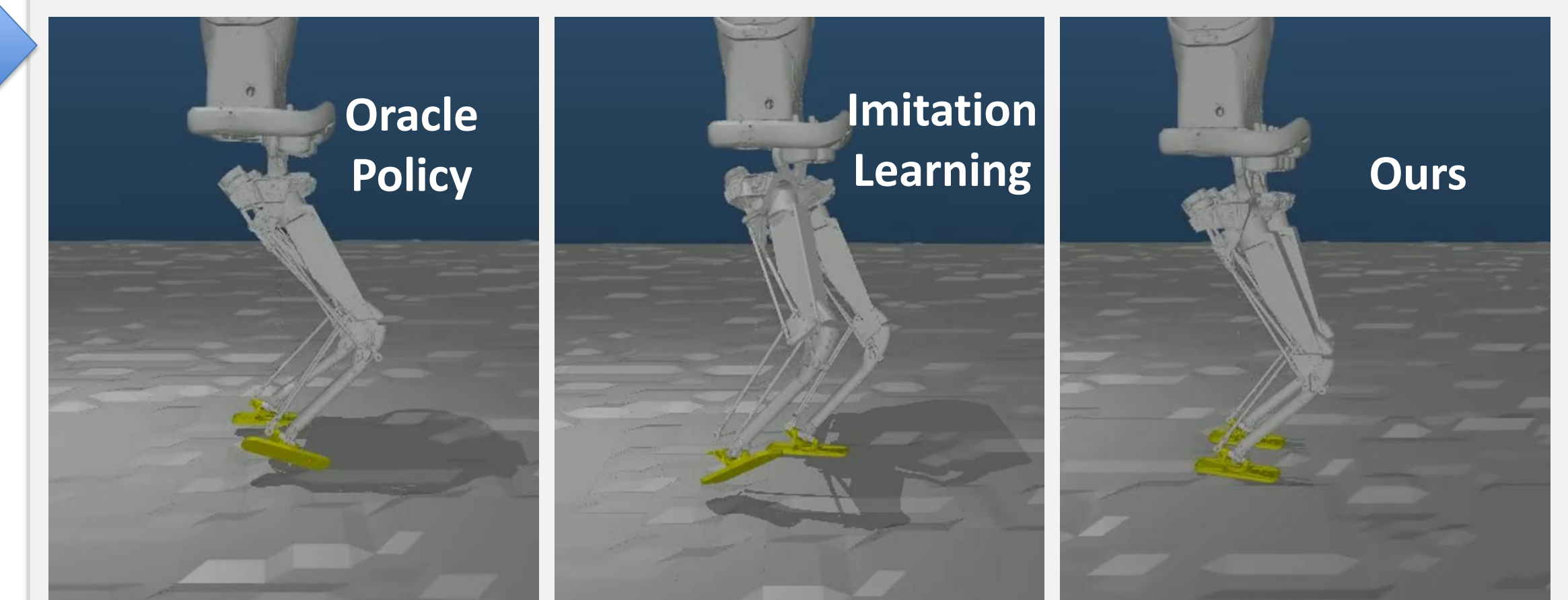
3 ROBUST OUTCOMES

Robust to increasing uncertainty (domain randomization level β)



Ours stays stable as uncertainty increases

Robust on uneven terrain



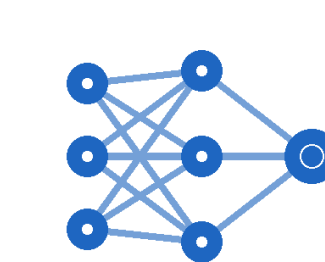
Qualitative takeaway

- Oracle / imitation can inherit failures on uneven terrain
- Residual RL allows exploration and deviation from the oracle
- Base + residual controller compensates realistic uncertainty

Robust across different robots



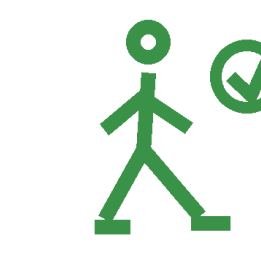
Model-based supervision + residual reinforcement learning enables robust torque-based locomotion under realistic uncertainties.



Guided
learning



Uncertainty
compensation



Robust
locomotion

